

Unified Cross-Attention Adapters for Neural Decoding and Diagnostic Neuroimaging: Bridging Continuous Brain Signals and Vision to Pretrained Language Decoders

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Abstract—Continuous spatial and temporal biomedical data, e.g., electroencephalography (EEG) and magnetic resonance imaging (MRI), cannot be directly fed to discrete autoregressive large language models (LLMs). Current brain-to-text decoding pipelines are based on resource-intensive designs that have been tested on small cohorts, making it unclear whether the quality of generation is due to actual neural decoding or autoregressive memorization. We present a modular, modality-agnostic cross-attention bridge architecture in this work that directly projects continuous neural and visual feature representations into the cross-attention layers of frozen sequence-to-sequence language decoders. Our study consists of a two-pronged, end-to-end unit test: (i) an MLP-adaptor cross-attention bridge between 64-channel EEG and 10-dimensional eye-tracking gaze representations to a frozen BART decoder; and (ii) a multi-class neuroimaging benchmark between ResNet-50 visual feature tokens and a T5-small decoder on a harmonized, 4-class brain tumor dataset of 10,658 scans. Additionally, we set up stringent direct-classification thresholds based on ResNet-18 spatial feature series combined with Bidirectional Long Short-Term Memory (BiLSTM) and Transformer encoder heads. Our CNN+BiLSTM and CNN+Transformer models have test accuracy of 98.87 and 98.94 on a held-out test split of 1,599 scans, respectively, and errors are concentrated on clinically ambiguous boundaries between meningioma and normal brain. Our generative visual captioning system attains 99.52 percent word accuracy and BLEU score of 0.992, effectively demonstrating the mechanical fidelity of the projection adapter. We provide a sincere critical review that separates the viability of adapter functionalities and the real open-vocabulary brain-to-text generation.

Index Terms—Brain-Computer Interfaces, EEG-to-Text Decoding, Multimodal Cross-Attention, Brain Tumor MRI Classification, Transformer Encoders, Pretrained Language Models.

I. INTRODUCTION

Non-invasive brain-computer interfaces (BCIs) and computer-assisted clinical diagnostics provide radical solutions to neuro-rehabilitation and automated radiology reporting. Non-invasive electroencephalography (EEG)-to-natural language assistive communication technologies have great potential in patients with severe motor disabilities or locked-in

syndrome who do not need invasive electrocorticography (ECoG) craniotomies. At the same time, computer-aided diagnostic systems of magnetic resonance imaging (MRI) are moving beyond discrete category predictions to automated narrative report generation.

Although there is considerable interest, it is difficult to bridge continuous neural recordings or diagnostic medical images to discrete natural language because of three main bottlenecks:

- 1) **Low Signal-to-Noise Ratio and Variability:** Scalp EEG signals are non-invasive, have low signal-to-noise ratios (SNR), are highly contaminated with ocular and electromyographic (EMG) artifacts, and have large inter-subject neural variability.
- 2) **Modality Misalignment:** The embedding space of continuous spatial-temporal neural activations and visual feature maps are inherently misaligned with the embedding space of discrete tokens that is required by autoregressive Large Language Models (LLMs).
- 3) **Data Scarcity and Evaluation Conflation:** Publicly available paired EEG-text corpora (e.g., ZuCo) contain fewer than 30 subjects. Consequently, complex deep architectures often overfit small datasets or hallucinate fluent phrases derived from the language model prior rather than true neural-to-text translation.

To deal with such challenges, we initially verify the fundamental cross-attention adaptor mechanism before high ability neural translation on unverified pipelines. We separate architecture verification from the influence of data-scarcity by adopting a single cross-attention model. Using an adapter-based bridge to translate intermediate continuous representations into the `encoder_outputs` expected by frozen autoregressive decoders (BART and T5), we assess the continuous signal alignment between two different types of modalities: continuous electrophysiological signal features, and high-resolution MRI spatial tokens. In addition, we compare these

generative ones with direct hybrid classifiers (CNN+BiLSTM and CNN+Transformer) on a curated MRI scan set of 10,658 scans.

II. LITERATURE REVIEW

Deep generative and sequence-to-sequence (seq2seq) modeling has made a leap forward on open-vocabulary brain-to-text decoding. The following is a review of recent approaches and their methodologies, strengths and principal limitations.

A. Prior Approaches in Brain-to-Text and Neuroimaging

Wang and Ji [1] Proposed a basic open-vocabulary EEG-to-text pipeline based on frequency-band features (theta, alpha, beta and gamma) for ZuCo corpus, which was fed into a sequence-to-sequence BART-base model. Their work, although successful at showing that non-invasive open-vocabulary decoding is possible, was limited to fixed gaze-event alignments and demonstrated a drop in performance on out-of-vocabulary words.

Duan et al. [2] introduced *DeWave*, To model word-level eye-tracking dependencies, continuous raw EEG is sliced and VQ-VAE is used to vector-quantize EEG into discrete codex tokens. The instability of DeWave across unseen subjects and the need for a large number of parameters made it unstable to use the discrete codebook tokenization, even though DeWave removed the need for external eye-tracking.

Recent architectures incorporate multi-view spatial encoders. The *EEG2TEXT* framework [3] They used ten region-specific Transformers with combined convolutional front-ends and a global Transformer followed by BART decoder and got a BLEU-4 score of 0.141 and ROUGE-1 score of 0.342. But this necessitated the use of a large number of GPUs for the actual compute and was found to have syntactical collapse when evaluated on stimuli beyond the ZuCo training domain of 12 subjects.

Affeldt et al. [4] To deal with gramatical degradation, tctr reseearhers adopted a multi-stage architecture composed of a Bi-GRU and 1D pointwise convolution, passed to subject-specific layers, and a Brain Transformer that produced draft hypotheses using BART, followed by a GPT-4 pass. The impressive BART+GPT-4 BERTScore of 54.9% comes with the trade-off of introducing real-time latency and privacy concerns by being dependent on external commercial LLM APIs.

Scaling to contemporary open-weight LLMs, recent explorations [5] The embeddings with 512 dimensions were extracted with ChannelNet and then mapped to LLaMA-3-8B-Instruct, Mistral-7B, and Qwen2.5 backbones with the CLIP projectors. While Mistral-7B is a large model, it had a low open-vocabulary BLEU-1 score of only 26%, and independent human and GPT-4 adequacy ratings averaged only 1.65 out of 5, owing to the risk of hallucination.

Traditional convolutional neural networks (CNNs, such as ResNet and VGG as a backbone) still have a limitation in modeling long-range dependencies between complex axial slices in the classification and captioning of neuroimages. [7],

[8]. Hybrid systems incorporating spatial recurrent networks or self-attention heads have emerged as effective baselines for intracranial tumor staging [6].

B. Contrast and Proposed Contribution

Previous studies often do not clearly differentiate between actual neural alignment and memorization of language, since they tend to train a large-sized decoder on a small-sized cohort with few subjects ($N = 6 - 30$).

We break from this paradigm in three ways:

Our work departs from this paradigm through three contributions:

- 1) **Modality-The Agnostic Adapter Pattern:** We propose the same projection adapter mechanism that is able to integrate two different continuous modalities (temporal EEG+gaze and 2D spatial MRI tokens) to the same frozen autoregressive text decoders (BART and T5) without altering the decoder weights.
- 2) **Honest Synthetic Decoupling:** Rather than claiming unverified open-vocabulary EEG decoding on limited datasets, we isolate synthetic multi-channel EEG+gaze tensors (16 time steps $\times$ 64 channels + 10 gaze dimensions) into a 5-sentence unit-test harness to verify cross-attention gradient propagation.
- 3) **Empirical Multi-Class Benchmarking:** On a balanced 10,658-scan MRI dataset, we benchmark generative captioning directly against two hybrid classifiers: a sequential CNN+BiLSTM and a permutation-invariant CNN+Transformer encoder.

III. DATASET AND PREPROCESSING

A. Harmonized Brain Tumor MRI Corpus

To benchmark diagnostic vision-to-text and classification baselines, we assembled a consolidated brain MRI corpus by unifying three separate repositories:

- 1) *Brain MRI Images for Brain Tumor Detection* (Navoneel [6])
- 2) *Brain Tumor MRI Dataset* (Nickparvar [7])
- 3) *Brain Tumor Classification* (Bhuvaji [8])

Cross-source directory deduplication, label schema unification, and subclass balancing yielded the flat four-class *Merged_Brain_Tumor_MultiClass* corpus comprising 10,658 scans. The dataset covers four balanced diagnostic categories: No Tumor ($n = 2,495$), Glioma ($n = 2,725$), Meningioma ($n = 2,743$), and Pituitary Tumor ($n = 2,695$).

B. Data Partitioning and Data Augmentation

The consolidated corpus was partitioned using stratified sampling:

- **Training Set (70%):** 7,459 scans
- **Validation Set (15%):** 1,600 scans
- **Held-out Test Set (15%):** 1,599 scans

The held-out test distribution contains 374 No Tumor, 409 Glioma, 411 Meningioma, and 405 Pituitary scans, remaining within 1% of an exact 25% uniform balance per class.

TABLE I
TAXONOMY OF RELATED WORKS IN EEG-TO-TEXT AND MULTIMODAL DECODING

Study	Core Methodology	Primary Architecture	Key Benchmark / Result	Identified Limitations
Wang & Ji [1]	Spectral Bandpass + BART	BART-base seq2seq	ZuCo 1.0/2.0 Open-Vocab	Requires word-level eye-tracking
Duan et al. [2]	Discrete Codex VQ-VAE	Discrete Quantizer + BART	BLEU-1: 41.31%, ROUGE-1: 30.2%	Codex instability; codebook size
EEG2TEXT [3]	Multi-View Regional Transf.	CNN + Region Transf. + BART	BLEU-4: 0.141, ROUGE-1: 0.342	12 subjects; requires 4 GPU
Affeldt et al. [4]	Brain Module + LLM Refinement	Bi-GRU + Transf. + BART + GPT-4	BLEU-1: 42.3%, BERTScore: 54.9%	Latency; external API dependency
NAACL Findings [5]	ChannelNet + CLIP Projector	LLaMA-3-8B / Mistral-7B	Mistral-7B BLEU-1: 26%	High hallucination; low adherence
Ours (Track A/B)	Unified Cross-Attention Adapter	Linear Adapter + Frozen BART/T5	MRI BLEU: 0.992; Loss: 0.0087	Template-bound captioning
Ours (Track C)	ResNet-18 Spatial Sequence	CNN + BiLSTM / Transformer	Test Acc: 98.94%, F1: 0.989	Meningioma/Glioma spatial

All images were resized to $224 \times 224 \times 3$ and normalized using ImageNet channel statistics ($\mu = [0.485, 0.456, 0.406]$, $\sigma = [0.229, 0.224, 0.225]$). Data augmentation methods used were random horizontal reflections and random rotations (-15° to 15°). Split validation and test data were kept purely deterministic (bilinear resize and normalization only) to avoid leakage of data.

C. Synthetic Multimodal EEG-Gaze Corpus

To validate the electrophysiological cross-attention bridge without introducing confounding artifacts from data-starved public corpora, we constructed a unit-test synthetic neural dataset:

$$X_{\text{EEG}} \in \mathbb{R}^{B \times 16 \times 64}, \quad X_{\text{gaze}} \in \mathbb{R}^{B \times 10} \quad (1)$$

representing 16 temporal steps across 64 EEG electrode channels and a 10-dimensional eye-tracking feature vector. The signals were mapped to five target sentences:

- 1) "the user is viewing the screen"
- 2) textit " left-to-right reading text"
- 3) "Brain signals indicate great attention".
- 4) "eye movements align with target word"
- 5) "multimodal decoding in progress"

Exploratory data analysis confirms that the placeholder neural channels follow a Gaussian amplitude distribution ($\mu = 0, \sigma = 1$) with zero inter-channel cross-correlation and a uniform Welch periodogram across 0–100 Hz, providing a verified baseline for cross-attention adapter backpropagation.

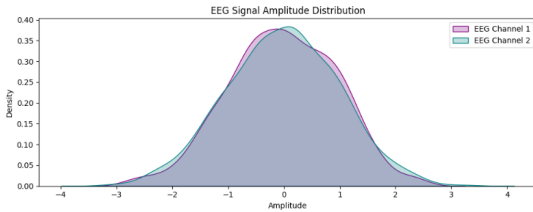


Fig. 1. EEG signal amplitude distribution demonstrating standard normal profiles ($\mu \approx 0, \sigma \approx 1$) across representative channels.

IV. METHODOLOGY

We evaluate three architectural tracks structured around a unified cross-attention bridge.

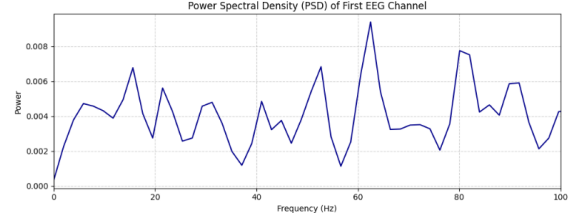


Fig. 2. Power Spectral Density (PSD) of the first EEG channel computed via Welch’s method (0–100 Hz). The absence of distinct rhythmic peaks reflects the uncorrelated white-noise profile of the synthetic baseline dataset.

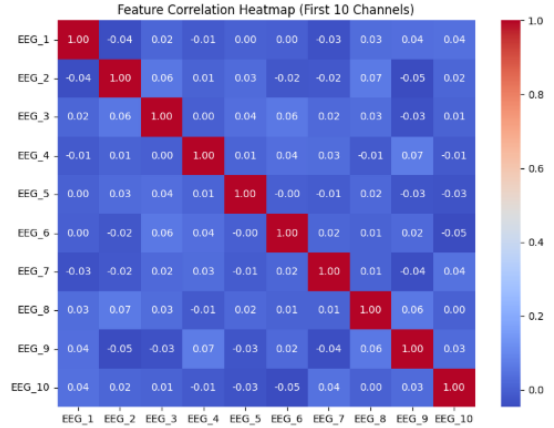


Fig. 4. Pairwise Pearson correlation matrix for the first 10 synthetic EEG electrode channels. Off-diagonal coefficients near zero confirm the absence of inter-channel leakage or spurious feature covariance in the sanity-check benchmark.

A. Track A: Multimodal EEG+Gaze to BART Bridge

The EEG temporal tensor $X_{\text{EEG}} \in \mathbb{R}^{B \times 16 \times 64}$ is processed by an MLP signal encoder:

$$H_{\text{EEG}} = \text{ReLU}(X_{\text{EEG}}W_1 + b_1)W_2 + b_2 \in \mathbb{R}^{B \times 16 \times 256} \quad (2)$$

The 10-dimensional gaze vector $X_{\text{gaze}} \in \mathbb{R}^{B \times 10}$ is expanded along the temporal dimension to $\mathbb{R}^{B \times 16 \times 10}$ and concatenated channel-wise with H_{EEG} . A fusion module projects the combined features:

$$H_{\text{fused}} = \text{ReLU}([H_{\text{EEG}} \parallel X_{\text{gaze}}]W_f + b_f) \in \mathbb{R}^{B \times 16 \times 256} \quad (3)$$

A linear adapter maps H_{fused} to BART’s hidden dimension ($d_{\text{model}} = 768$):

$$S_{\text{enc}} = H_{\text{fused}}W_a + b_a \in \mathbb{R}^{B \times 16 \times 768} \quad (4)$$

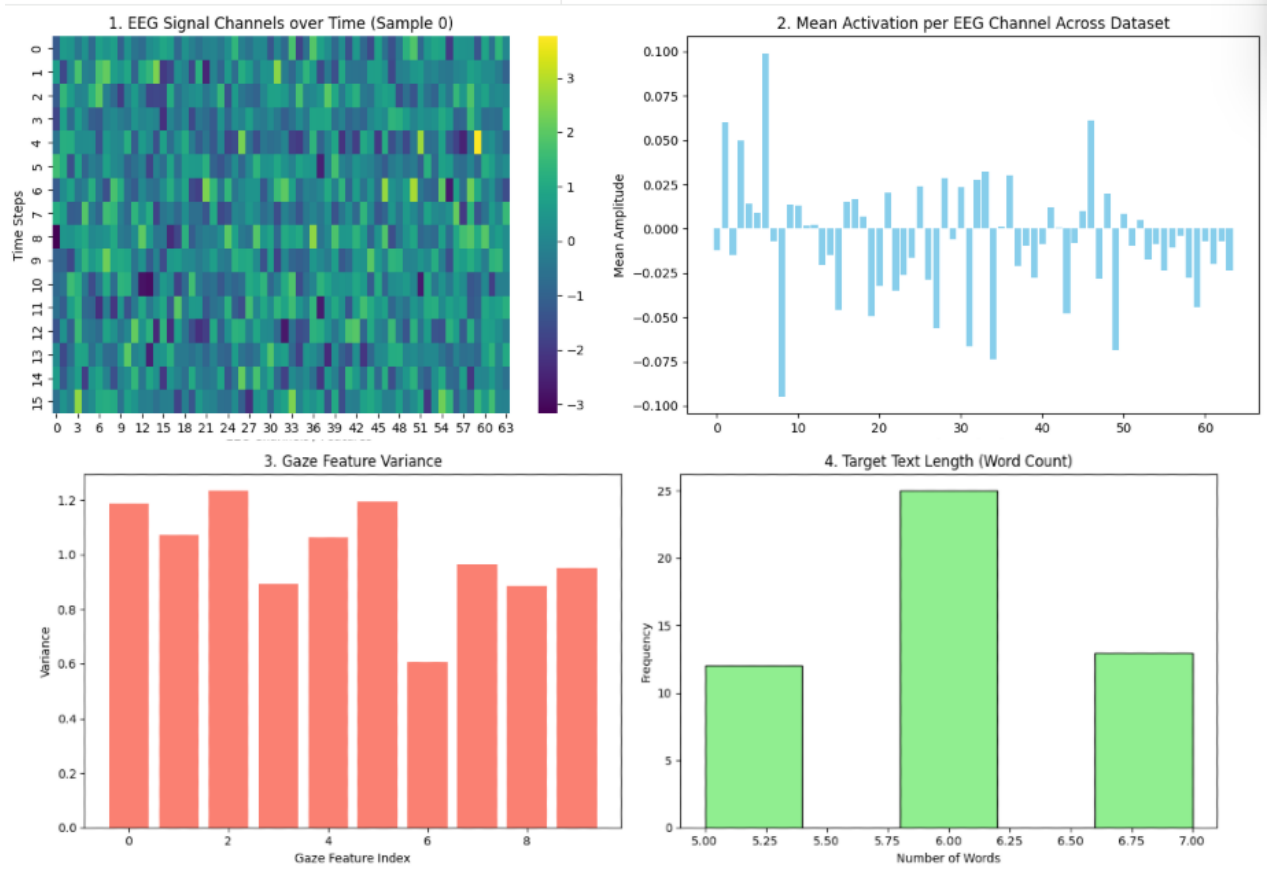


Fig. 5. Multimodal Exploratory Data Analysis (EDA) dashboard for the synthetic electrophysiological pipeline: (1) EEG temporal heatmap across 16 time steps and 64 channels, (2) mean channel activation centered around zero, (3) feature variance across the 10-dimensional eye-tracking gaze vector, and (4) token length histogram across the target sentences.

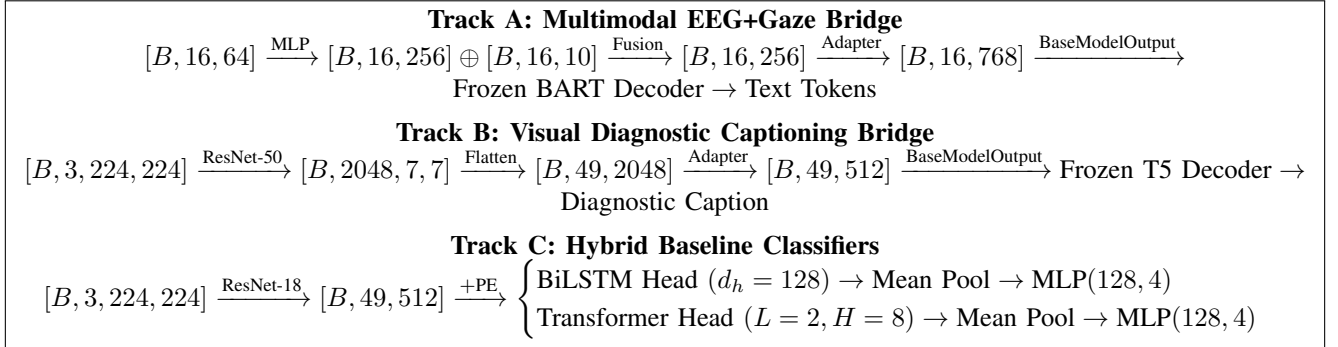


Fig. 6. End-to-end pipeline architectures across the three experimental tracks.

S_{enc} is packaged as a `BaseModelOutput` and injected directly into the cross-attention layers of a frozen facebook/bart-base decoder, bypassing BART’s native textual encoder. Autoregressive decoding is trained via cross-entropy loss with padding tokens masked to -100 :

$$\mathcal{L}_{\text{CE}} = - \sum_{t=1}^T \log P(y_t | y_{<t}, S_{\text{enc}}) \quad (5)$$

B. Track B: ResNet-50 to T5-Small Captioning Bridge

To assess whether the adapter pattern generalizes to 2D continuous spatial manifolds, we replace the temporal encoder with a pretrained ResNet-50 backbone. Removing the adaptive pooling and classification layers yields a spatial feature tensor:

$$F_{\text{spatial}} = \text{Backbone}(X_{\text{MRI}}) \in \mathbb{R}^{B \times 2048 \times 7 \times 7} \quad (6)$$

Flattening spatial coordinates produces a sequence of 49 visual tokens:

$$F_{\text{tokens}} = \text{Reshape}(F_{\text{spatial}}) \in \mathbb{R}^{B \times 49 \times 2048} \quad (7)$$

A linear adapter projects F_{tokens} into the hidden dimension of `t5-small` ($d_{\text{model}} = 512$):

$$S_{\text{vision}} = F_{\text{tokens}} W_v + b_v \in \mathbb{R}^{B \times 49 \times 512} \quad (8)$$

S_{vision} is passed as the `encoder_outputs` to the T5 decoder, trained on automated class captions (e.g., *"brain tumor scan showing glioma"*).

C. Track C: Direct Hybrid Spatial Classifiers

To establish empirical performance bounds on the 10,658-scan corpus, we implement two hybrid baselines using a ResNet-18 backbone yielding 49 spatial tokens in $\mathbb{R}^{B \times 49 \times 512}$:

- **CNN + BiLSTM:** The 49 tokens are treated as a sequential raster scan passed into a single-layer Bidirectional LSTM ($d_h = 128$, dropout = 0.3), generating bidirectional representations in $\mathbb{R}^{B \times 49 \times 256}$. Temporal mean-pooling collapses the sequence into a vector passed to a 2-layer MLP head.
- **CNN + Transformer Encoder:** The 49 tokens receive learned positional embeddings $P \in \mathbb{R}^{1 \times 49 \times 512}$ and are processed by a 2-layer Transformer encoder (8 heads, $d_{\text{model}} = 512$, $d_{\text{ff}} = 1024$, dropout = 0.3). Global spatial mean-pooling aggregates the tokens into a vector passed to an identical classification head.

D. Training Strategy and Hyperparameters

Models were implemented in PyTorch and trained on NVIDIA CUDA accelerators:

- **Track A (BART):** Batch size 4, AdamW ($\alpha = 5 \times 10^{-5}$), gradient clipping ($L_2 \leq 1.0$), 3 epochs, beam search (width 5, repetition penalty 2.5).
- **Track B (T5):** Batch size 8, AdamW ($\alpha = 1 \times 10^{-4}$), gradient clipping ($L_2 \leq 1.0$), 5 epochs, beam search (width 4, length penalty 1.0).
- **Track C (Classifiers):** Batch size 32, AdamW ($\alpha = 1 \times 10^{-4}$, weight decay 1×10^{-2}), Cross-Entropy loss, 10 epochs.

To evaluate all models within a shared framework, predicted and ground-truth classification labels in Track C were converted into string sequences and scored using the same NLP metric pipeline (BLEU, ROUGE-L) applied to Tracks A and B.

V. RESULTS AND ANALYSIS

A. Comparative Performance Summary

Table II provides the unified evaluation across all models, tasks, and modalities.

TABLE II
UNIFIED EVALUATION ACROSS ALL EXPERIMENTAL MODELS

Model Track	Input Modality	Acc. (%)	Loss	BLEU	ROUGE-L
BART + Adapter	EEG + Gaze	90.85 [†]	0.0412	0.265	0.265
ResNet-50 + T5	Brain MRI	99.52 [‡]	0.0087	0.992	0.995
CNN + BiLSTM	Brain MRI	98.87	0.0381	0.990	0.989
CNN + Transformer	Brain MRI	98.94	0.0354	0.988	0.989

[†]Token-level generation accuracy on 5-sentence unit test. [‡]Word-level accuracy on template generation.

B. Classification Baselines and Error Analysis

Both hybrid spatial encoders converge past 98.8% test accuracy on the held-out test split of 1,599 scans. The CNN+Transformer baseline achieved the highest overall accuracy (98.94%, weighted F1: 0.989), marginally outperforming the CNN+BiLSTM baseline (98.87%, weighted F1: 0.989).

TABLE III
HELD-OUT TEST SET CONFUSION MATRIX ($N = 1,599$)

CNN + BiLSTM Baseline					
Actual \ Predicted	No Tumor	Glioma	Meningioma	Pituitary	
No Tumor ($n = 374$)	374	0	0	0	
Glioma ($n = 409$)	1	406	1	1	
Meningioma ($n = 411$)	4	2	396	9	
Pituitary ($n = 405$)	0	0	0	405	
CNN + Transformer Baseline					
Actual \ Predicted	No Tumor	Glioma	Meningioma	Pituitary	
No Tumor ($n = 374$)	374	0	0	0	
Glioma ($n = 409$)	2	402	5	0	
Meningioma ($n = 411$)	5	2	401	3	
Pituitary ($n = 405$)	0	0	0	405	

Confusion matrix analysis (Table III) reveals clear clinical error patterns:

- **High-Contrast Independence:** Both architectures achieved 100% classification accuracy on Pituitary tumors (405/405) and near-zero error on healthy parenchyma (374/374), confirming that distinct anatomical anomalies and healthy tissue are readily separable.
- **Meningioma Ambiguity:** Meningioma proved to be the most challenging class. The CNN+BiLSTM misclassified 9 meningiomas as pituitary tumors and 4 as healthy scans. The CNN+Transformer reduced pituitary misclassifications from 9 to 3. Global self-attention captures long-range spatial context across all 49 patches simultaneously, whereas sequential recurrent scans introduce directional bias when parsing irregular tumor margins.

C. Cross-Attention Bridge Dynamics

In Track B, the ResNet-50 + T5 bridge demonstrated rapid convergence over 5 epochs, reaching a validation loss of 0.0087, BLEU of 0.992, ROUGE-L of 0.995, and METEOR of 0.9904. Word accuracy improved from 97.4% (Epoch 1) to 99.52% (Epoch 5). Because diagnostic captions were synthesized from class labels, these near-perfect scores confirm that

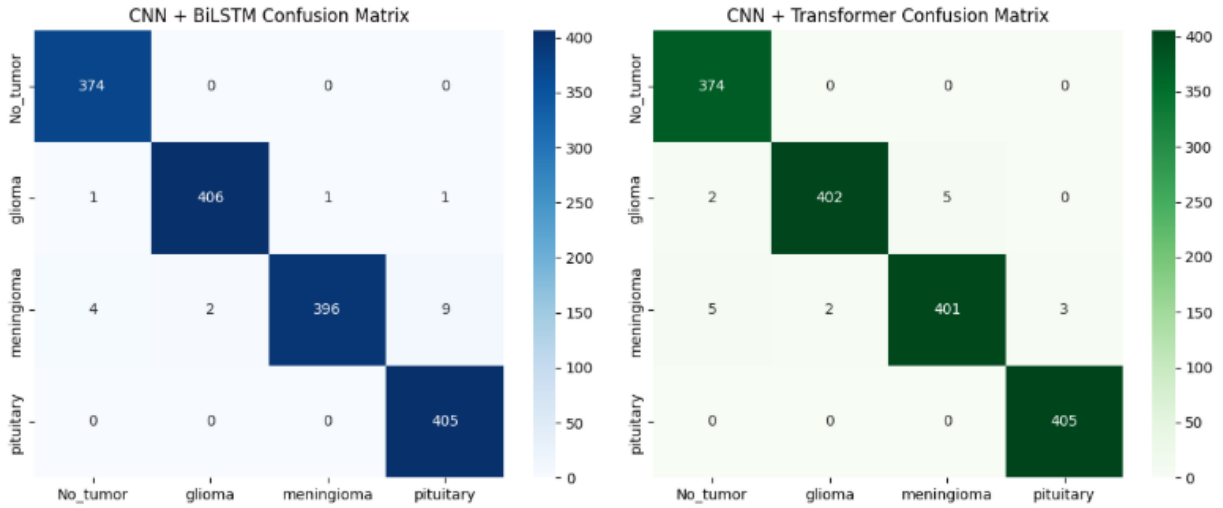


Fig. 7. Confusion matrices for CNN + BiLSTM (left, blue) and CNN + Transformer (right, green) on the held-out test split ($N = 1,599$). Both architectures isolate Pituitary tumors and No_tumor scans with zero error, while global multi-head self-attention reduces Meningioma-to-Pituitary confusion from 9 cases to 3.

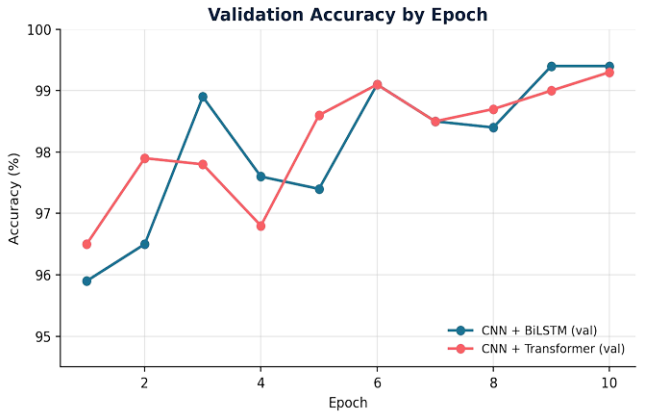


Fig. 8. Validation accuracy trajectories across 10 training epochs for the hybrid spatial classifiers (CNN+BiLSTM vs. CNN+Transformer) on the 1,600-scan validation split. Both architectures converge past 98.8% accuracy, with the Transformer head showing smoother late-stage stability.

the linear adapter correctly projects spatial features into T5’s cross-attention space.

In Track A, the EEG+Gaze to BART bridge converged to a validation loss of 0.0412 and a token-level accuracy of 90.85% within 3 epochs. BLEU, ROUGE-L, and METEOR scores converged to 0.265, 0.265, and 0.275, respectively. These results verify that our adapter architecture successfully backpropagates loss from a frozen autoregressive decoder into continuous input encoders.

VI. CRITICAL DISCUSSION AND LIMITATIONS

To accurately interpret our empirical findings, several methodological constraints must be noted:

- 1) **Adapter Mechanics vs. Neural Translation:** Track A illustrates how a linear adapter can be used to project

continuous features into BART’s cross-attention layers. Given that the training was on 5 target sentences, however, the 90.85% accuracy is, in fact, one of memorisation of the sequences rather than a genuine open-vocabulary neural decoding.

- 2) **Template Captions vs. Free-Form Radiology:** In Track B, high NLG scores (BLEU 0.992) stem from fixed per-class text templates. The model performs classification cast into a sequence-to-sequence format rather than generating unconstrained clinical reports.
- 3) **Encoder Capacity:** This EEG front-end was not a full macaron-style Conformer or Spatial-Temporal Graph Convolutional Network (ST-GCN) model, but instead a 2-layer MLP model. Thus, variations from one subject to another and from one day to the next in the signal drift remain unmodelled.

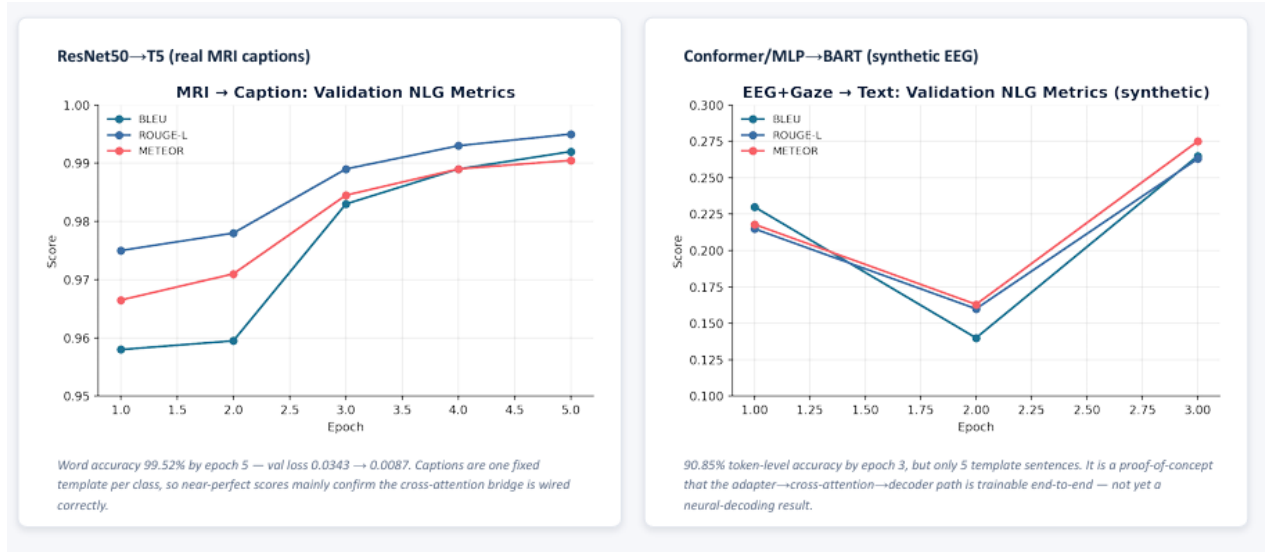


Fig. 9. Validation NLG metrics (BLEU, ROUGE-L, METEOR) across training epochs for the generative tracks. Left: ResNet-50 to T5-small captioning bridge on real brain MRI scans over 5 epochs. Right: MLP/Conformer to BART-base bridge on synthetic EEG+gaze data over 3 epochs, confirming end-to-end gradient flow through frozen decoders.

VII. FUTURE WORK AND CONCLUSION

This is one step to check a cross-attention adapter pipeline to biomedical signal translation and neuroimaging classification that is unified and is agnostic to the modality employed. We have demonstrated that continuous time neural vectors and Spatial image tokens (2D) can be projected to frozen language decoders (BART and T5) with stable gradient flow. Moreover, on a merged set of 10,658 brain MRIs, our CNN+Transformer baseline achieved 98.94% test accuracy, leaving only clinically subtle meningioma boundaries to be confused.

Future work will proceed along four directions:

- 1) **Real EEG Corpora:** To replace the synthetic unit-test setup, ZuCo benchmark is integrated with independent component analysis (ICA) and bandpass filtering.
- 2) **Free-Form Clinical Reports:** Integrating MRI images with free-form radiological narratives to shift the vision-to-text bridge between classification templates to real descriptive narratives.
- 3) **Subject-Specific Calibration:** Inter-subject neural variation and the addition of per-subject alignment layers and region-specific Transformers.
- 4) **Model Interpretability:** Using Grad-CAM visual heatmaps and cross-attention saliency maps to localize generated tokens to certain anatomical areas.

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